

Network Monitoring and Wireshark

CSE 322

Acknowledgement

- <https://studylib.net/doc/17803129/cap6135--malware-and-software-vulnerability-analysis-netw...>
- <https://slideplayer.com/slide/5692294/>
- <http://ilta.ebiz.uapps.net/ProductFiles/productfiles/672/wireshark.ppt>
- UC Berkley course “EE 122: Intro to Communication Networks”
 - <http://www.eecs.berkeley.edu/~jortiz/courses/ee122/presentations/Wireshark.ppt>
- Other resources:
 - http://openmaniak.com/wireshark_filters.php

Motivation for Network Monitoring

- Essential for Network Management
 - Router and Firewall policy
 - Detecting abnormal/error in networking
 - Access control
- Security Management
 - Detecting abnormal traffic
 - Traffic log for future forensic analysis

Tools Overview

- Tcpdump
 - Unix-based command-line tool used to intercept packets
 - Including *filtering* just the packets of interest
 - Reads “live traffic” from interface specified using *-i* option ...
 - ... or from a previously recorded *trace file* specified using *-r* option
 - You create these when capturing live traffic using *-w* option
- Tshark
 - Tcpdump-like capture program
 - Very similar behavior and flags to tcpdump
- Wireshark
 - GUI for displaying tcpdump/tshark packet traces

tcpdump Example

- Ran tcpdump on a Unix machine
- Example: first few lines of the output

```
01:46:28.808262 IP danjo.CS.Berkeley.EDU.ssh >  
    adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481: .  
    2513546054:2513547434(1380) ack 1268355216 win 12816  
01:46:28.808271 IP danjo.CS.Berkeley.EDU.ssh >  
    adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481: P 1380:2128(748) ack 1  
    win 12816  
01:46:28.808276 IP danjo.CS.Berkeley.EDU.ssh >  
    adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481: . 2128:3508(1380) ack 1  
    win 12816  
01:46:28.890021 IP adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481 >  
    danjo.CS.Berkeley.EDU.ssh: P 1:49(48) ack 1380 win 16560
```

What Does a Line Convey?

Timestamp This Source Port number (22)

01:46:28.808262 IP danjo.CS.Berkeley.EDU ssh >
adsl-69-228-230-7.dsl.pltn13.pacbell.net 2481:..

2513546054:2513547434(1380) ack 1268355216 win 12816

Destination host name

Destination port number

TCP specific information

- Different output formats for different packet types

Similar Output from Tshark

```
1190003744.940437 61.184.241.230 -> 128.32.48.169 SSH
  Encrypted request packet len=48
1190003744.940916 128.32.48.169 -> 61.184.241.230 SSH
  Encrypted response packet len=48
1190003744.955764 61.184.241.230 -> 128.32.48.169 TCP 6943
  > ssh [ACK] Seq=48 Ack=48 Win=65514 Len=0 TSV=445871583
  TSER=632535493
1190003745.035678 61.184.241.230 -> 128.32.48.169 SSH
  Encrypted request packet len=48
1190003745.036004 128.32.48.169 -> 61.184.241.230 SSH
  Encrypted response packet len=48
1190003745.050970 61.184.241.230 -> 128.32.48.169 TCP 6943
  > ssh [ACK] Seq=96 Ack=96 Win=65514 Len=0 TSV=445871583
  TSER=632535502
```


Demo 1 – Basic Run

- Syntax:

tcpdump [options] [filter expression]

- To do -
 - \$ sudo tcpdump -i eth0
 - Sudo command allows you to run tcpdump in root privilege
 - On your own Unix machine, you can run it using “sudo” or directly run “tcpdump” if you have root privilege
- Observe the output

Filters

- We are often not interested in all packets flowing through the network
- Use filters to capture only packets of interest to us

Demo 2

1. Capture only udp packets
 - `tcpdump "udp"`
2. Capture only tcp packets
 - `tcpdump "tcp"`

Demo 2 (contd.)

1. Capture only UDP packets with destination port 53 (DNS requests)
 - `tcpdump "udp dst port 53"`
2. Capture only UDP packets with source port 53 (DNS replies)
 - `tcpdump "udp src port 53"`
3. Capture only UDP packets with source or destination port 53 (DNS requests and replies)
 - `tcpdump "udp port 53"`

Running tcpdump

- Requires superuser/administrator privileges on Unix
 - <http://www.tcpdump.org/>
 - You can do it on your own Unix machine
 - You can install a Linux OS in Vmware on your windows machine
- Tcpdump for Windows
 - WinDump: <http://www.winpcap.org/windump/>
 - Free software
 - Might have compatibility issues

So What is WireShark?

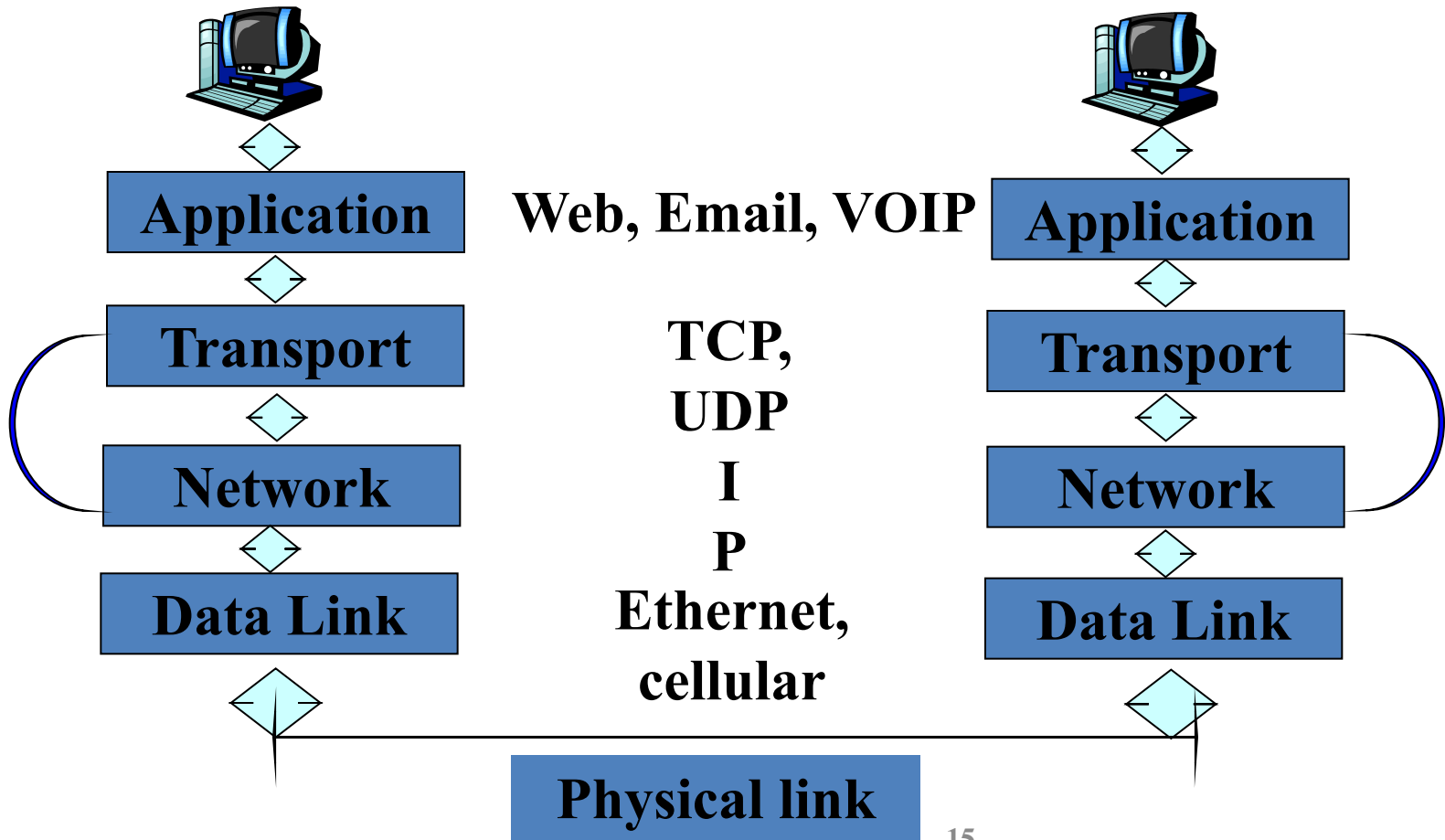
- Packet sniffer/protocol analyzer
- Open Source Network Tool
- Latest version of the ethereal tool



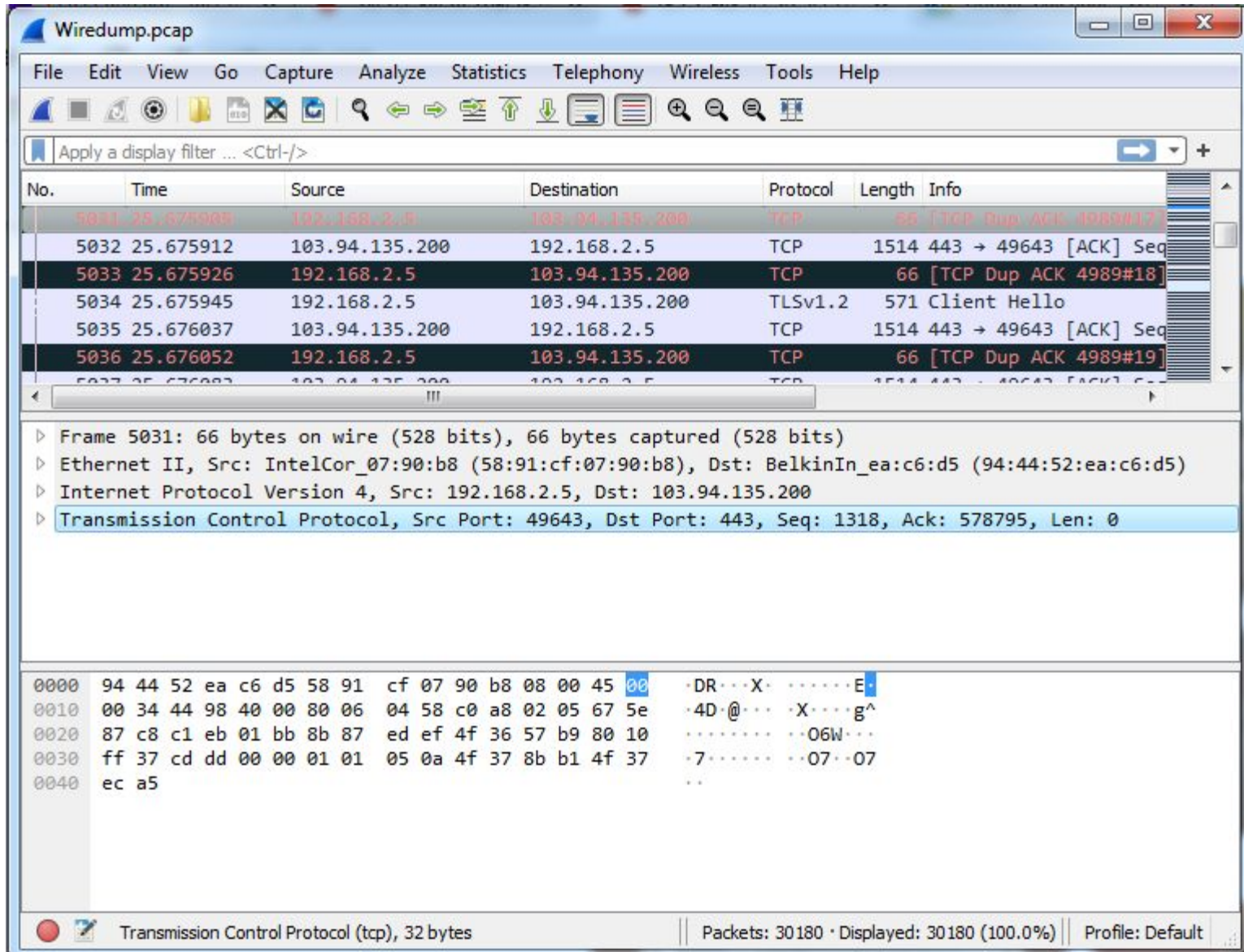
#	Wireshark	Tcpdump
1	A GUI tool to catch data packets	A CLI-based packet capturing tool
2	It does packet analysis, decode data payloads if the encryption keys are identified, and recognize data payloads from file transfers such as smtp, http, etc.	Tcpdump only provides do a simple analysis of such types of traffic, such as DNS queries
3	It has advanced network interfaces	It has conventional interfaces
4	Good for complex filters	Used for simple filters
5	It provides decoding of protocol-based packet capturing	It is less efficient in decoding compared to Wireshark

Scope of WireShark

- Internet and all ...



Wireshark Interface



Wireshark Interface – Different Parts

command menus

display filter specification

listing of captured packets

details of selected packet header

packet content in hexadecimal and ASCII

No.	Time	Source	Destination	Protocol	Length	Info
5032	25.675912	103.94.135.200	192.168.2.5	TCP	1514	443 → 49643 [ACK] Seq=...
5033	25.675926	192.168.2.5	103.94.135.200	TCP	66	[TCP Dup ACK 4989#18]
5034	25.675945	192.168.2.5	103.94.135.200	TLSv1.2	571	Client Hello
5035	25.676037	103.94.135.200	192.168.2.5	TCP	1514	443 → 49643 [ACK] Seq=...

Frame 5031: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)

Ethernet II, Src: IntelCor_07:90:b8 (58:91:cf:07:90:b8), Dst: BelkinIn_ea:c6:d5 (94:44:52:ea:c6:d5)

Internet Protocol Version 4, Src: 192.168.2.5, Dst: 103.94.135.200

Transmission Control Protocol, Src Port: 49643, Dst Port: 443, Seq: 1318, Ack: 578795, Len: 0

```
0000  94 44 52 ea c6 d5 58 91 cf 07 90 b8 08 00 45 00  DR...X...E
0010  00 34 44 98 40 00 80 06 04 58 c0 a8 02 05 67 5e  4D@...X...g^
0020  87 c8 c1 eb 01 bb 8b 87 ed ef 4f 36 57 b9 80 10  .....06W...
0030  ff 37 cd dd 00 00 01 01 05 0a 4f 37 8b b1 4f 37  7.....0707
0040  ec a5
```

Transmission Control Protocol (tcp), 32 bytes

Packets: 30180 · Displayed: 30180 (100.0%)

Profile: Default

Status Bar

The image shows a Wireshark window titled "Wiredump.pcap". The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar with various icons. A display filter bar at the top right contains the text "Apply a display filter ... <Ctrl-/>".

The main packet list table displays the following data:

No.	Time	Source	Destination	Protocol	Length	Info
5031	25.675905	192.168.2.5	103.94.135.200	TCP	66	[TCP Dup ACK 4989#17]
5032	25.675912	103.94.135.200	192.168.2.5	TCP	1514	443 → 49643 [ACK] Seq
5033	25.675926	192.168.2.5	103.94.135.200	TCP	66	[TCP Dup ACK 4989#18]
5034	25.675945	192.168.2.5	103.94.135.200	TLSv1.2	571	Client Hello
5035	25.676037	103.94.135.200	192.168.2.5	TCP	1514	443 → 49643 [ACK] Seq

Below the packet list, the packet details pane shows the following information for the selected packet (Frame 5031):

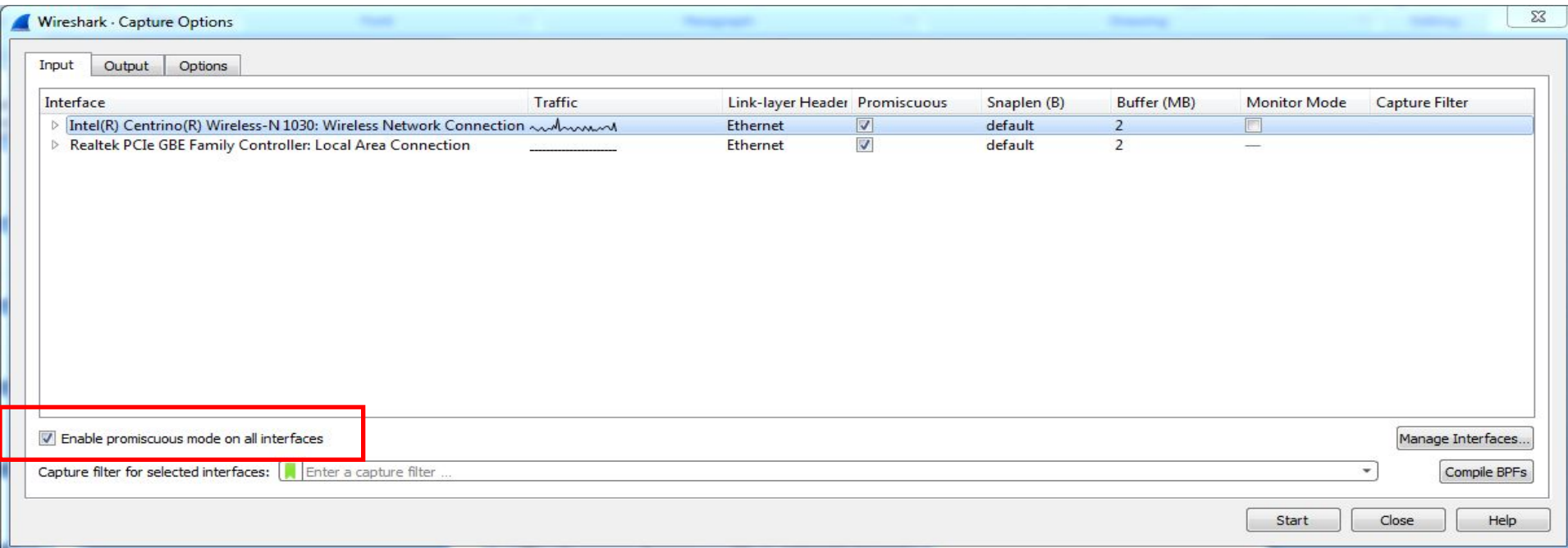
- Frame 5031: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
- Ethernet II, Src: IntelCor_07:90:b8 (58:91:cf:07:90:b8), Dst: BelkinIn_ea:c6:d5 (94:44:52:ea:c6:d5)
- Internet Protocol Version 4, Src: 192.168.2.5, Dst: 103.94.135.200
- Transmission Control Protocol, Src Port: 49643, Dst Port: 443, Seq: 1318, Ack: 578795, Len: 0

The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII. The first few bytes are highlighted in blue: 94 44 52 ea c6 d5 58 91 cf 07 90 b8 08 00 45 00. The ASCII representation shows the start of an Ethernet II frame header.

The status bar at the bottom of the window displays the following information:

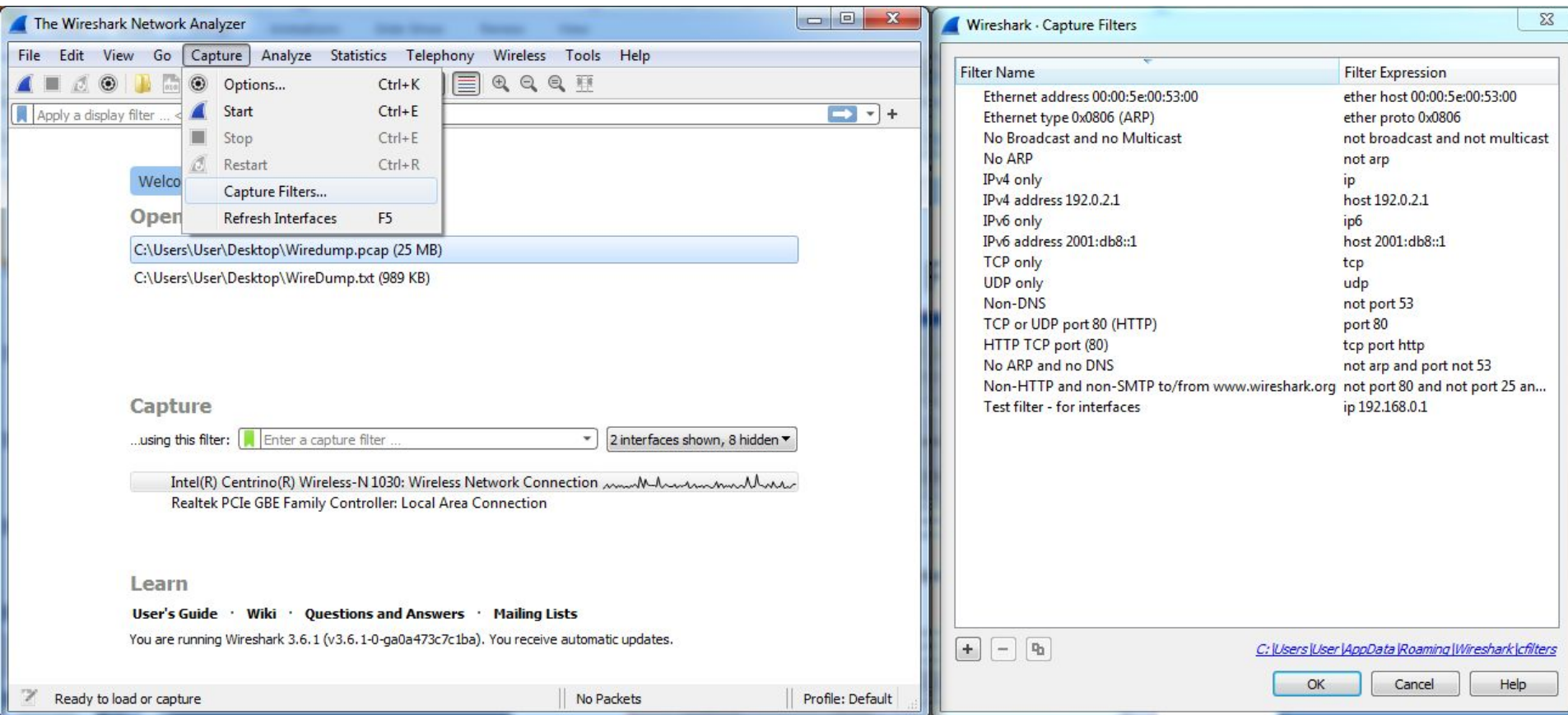
- Transmission Control Protocol (tcp), 32 bytes
- Packets: 30180 · Displayed: 30180 (100.0%)
- Profile: Default

Capture Options



- Promiscuous mode is used to capture all traffic
- Sometime this does not work
 - Driver does not support
 - You are on a switched LAN

Capture Filter



Capture Filter Examples

host 10.1.11.24

host 192.168.0.1 and host 10.1.11.1

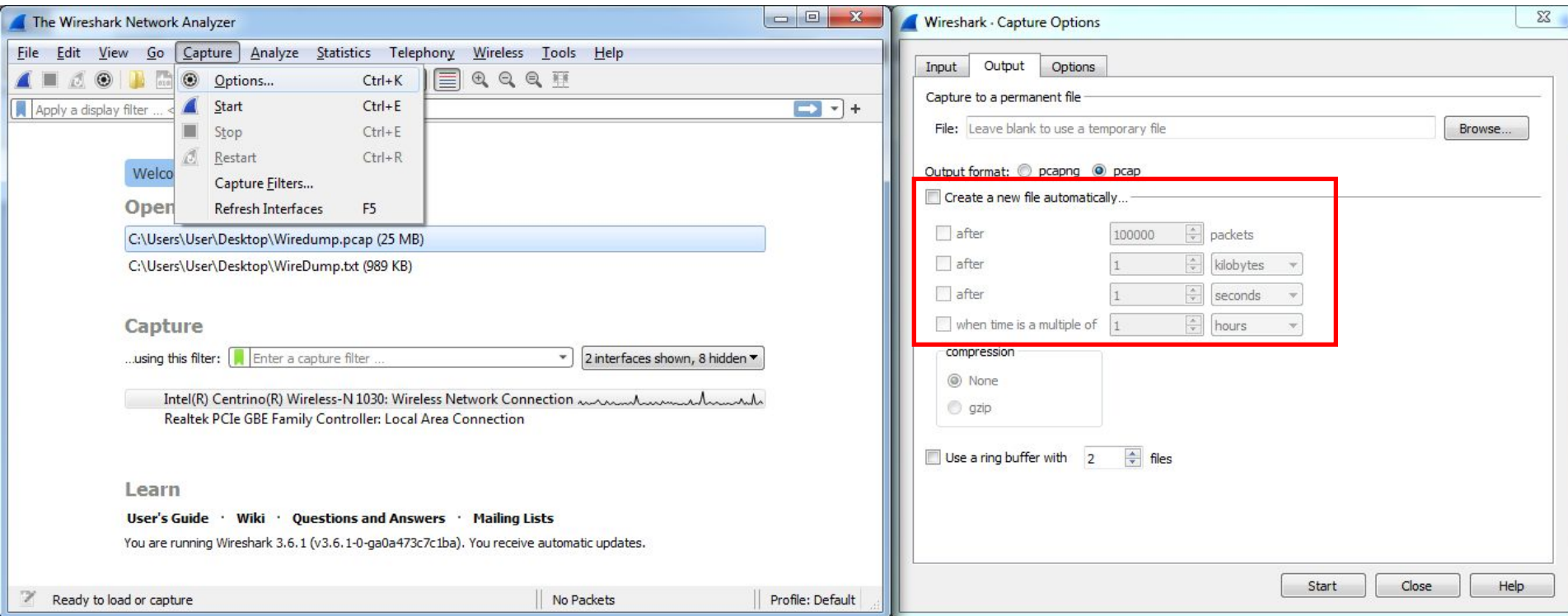
tcp port http

ip

not broadcast not multicast

ether host 00:04:13:00:09:a3

Capture Options



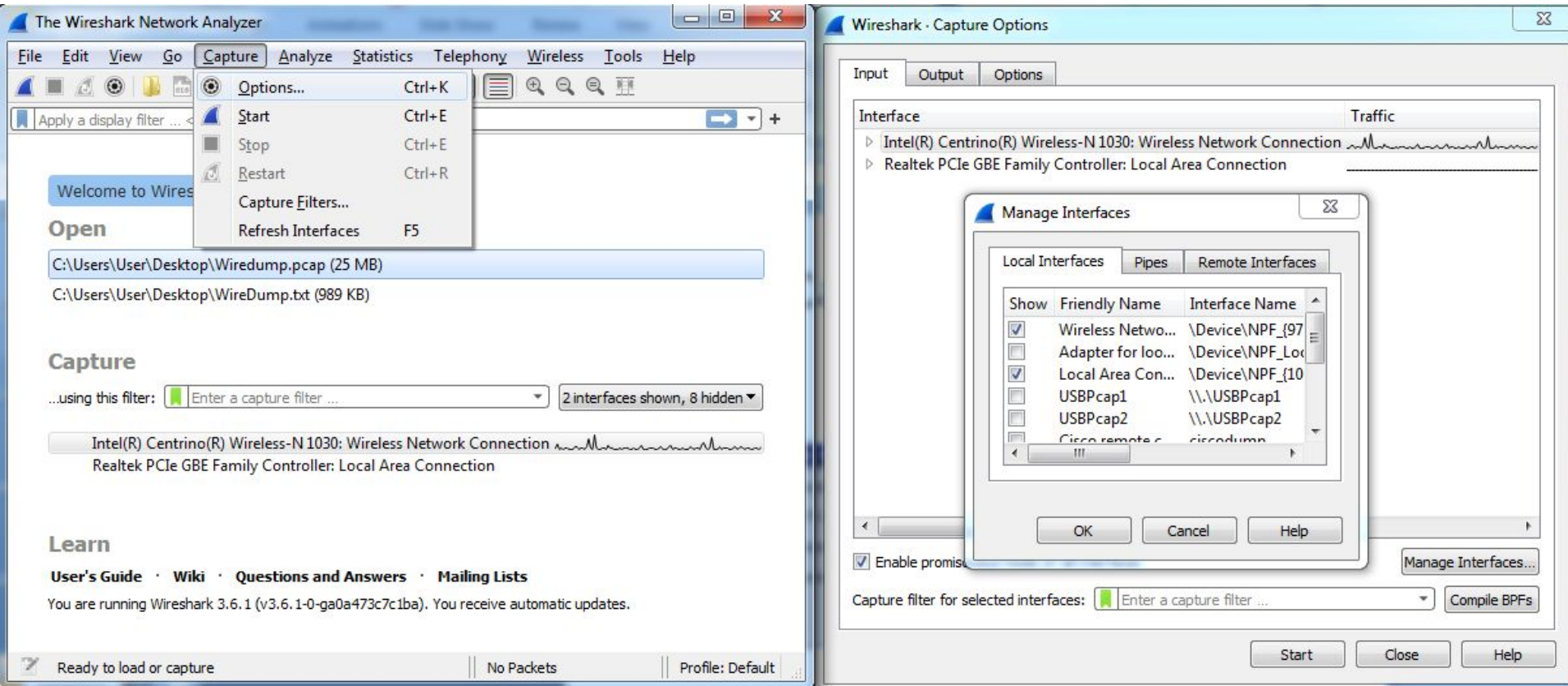
- Ring buffer

- Addresses a common issue many analysts encounter when capturing packets - huge traces
- When you use a Ring Buffer, you can define how many files you want to capture and various parameters that affects the file size (# of packets, bytes, and time)

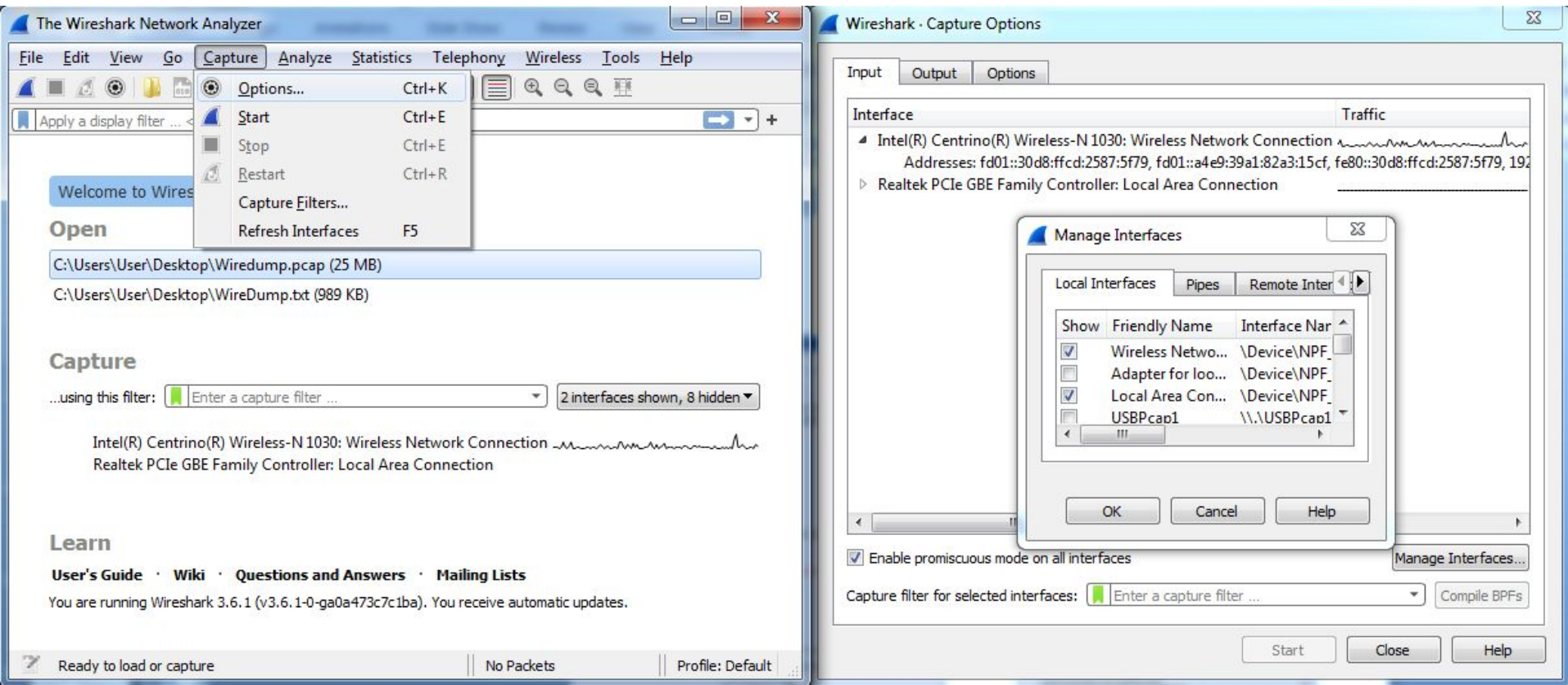
- Link:

<https://www.networkcomputing.com/networking/working-ring-buffer-wireshark-files>

Capture Interfaces



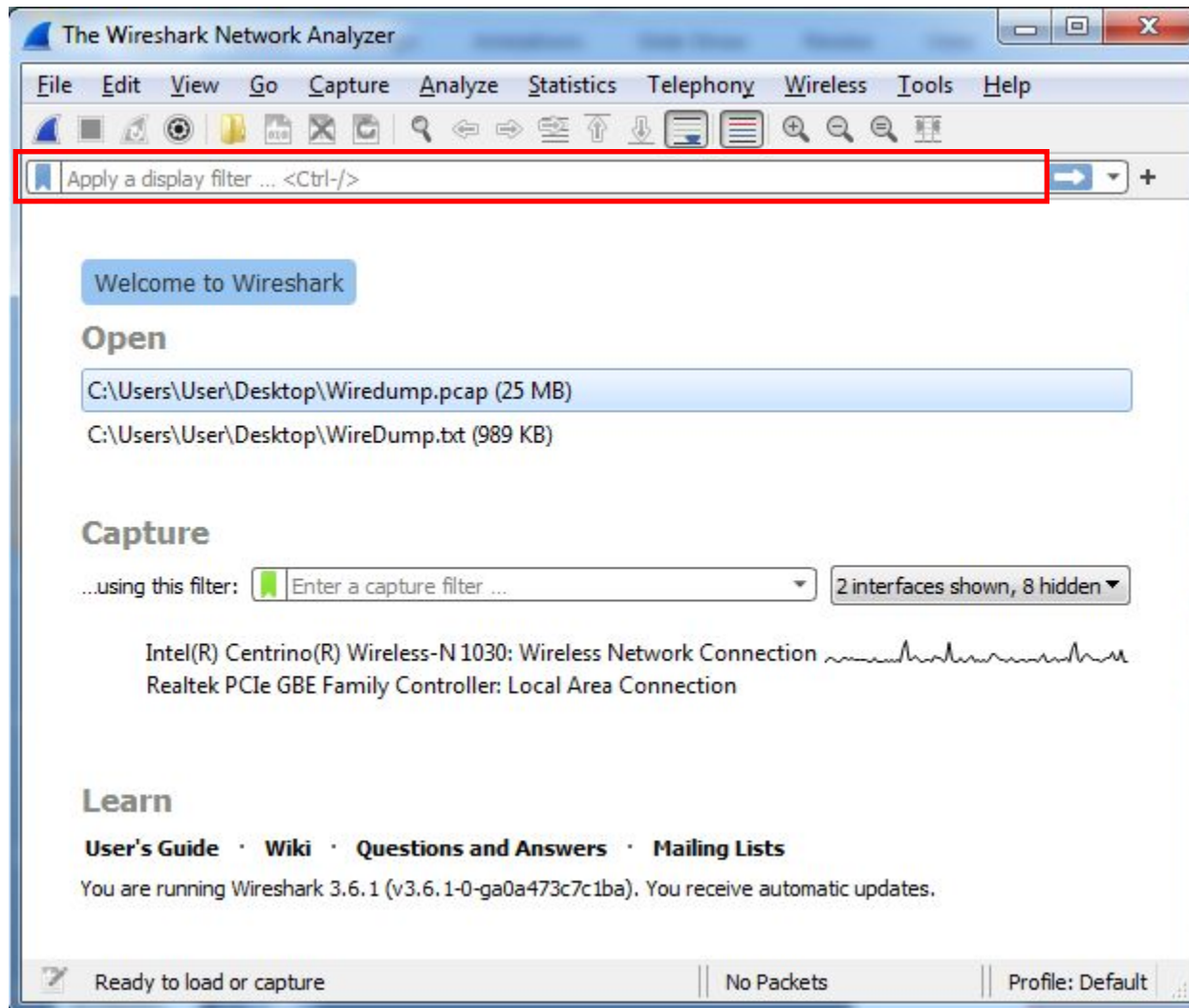
Interface Details



Display Filters (Post-Filters)

- Display filters (also called post-filters) only filter the view of what you are seeing
 - In analysis
 - All packets in the capture still exist in the trace
- Display filters use their own format and are much more powerful than capture filters

Display Filter



Display Filter Examples

```
ip.src==10.1.11.00/24
```

```
ip.addr==192.168.1.10 && ip.addr==192.168.1.20
```

```
tcp.port==80 || tcp.port==3389
```

```
!(ip.addr==192.168.1.10 && ip.addr==192.168.1.20)
```

```
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) &&  
  (tcp.port==445 || tcp.port==139)
```

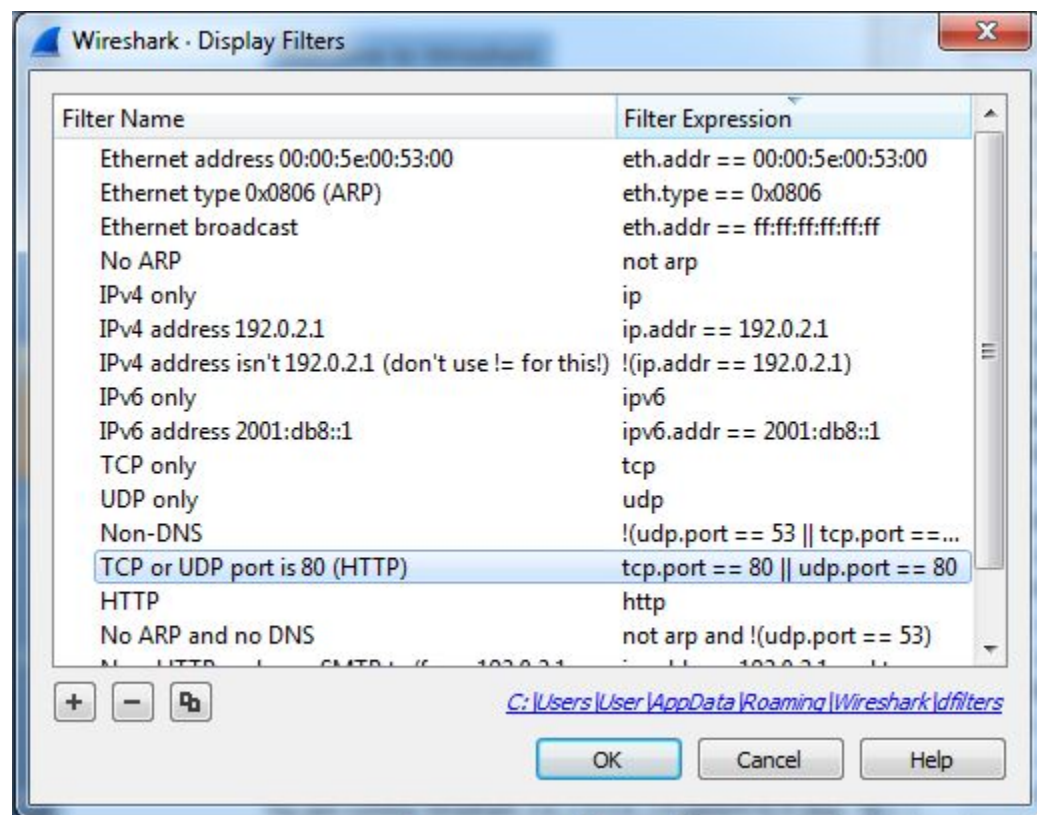
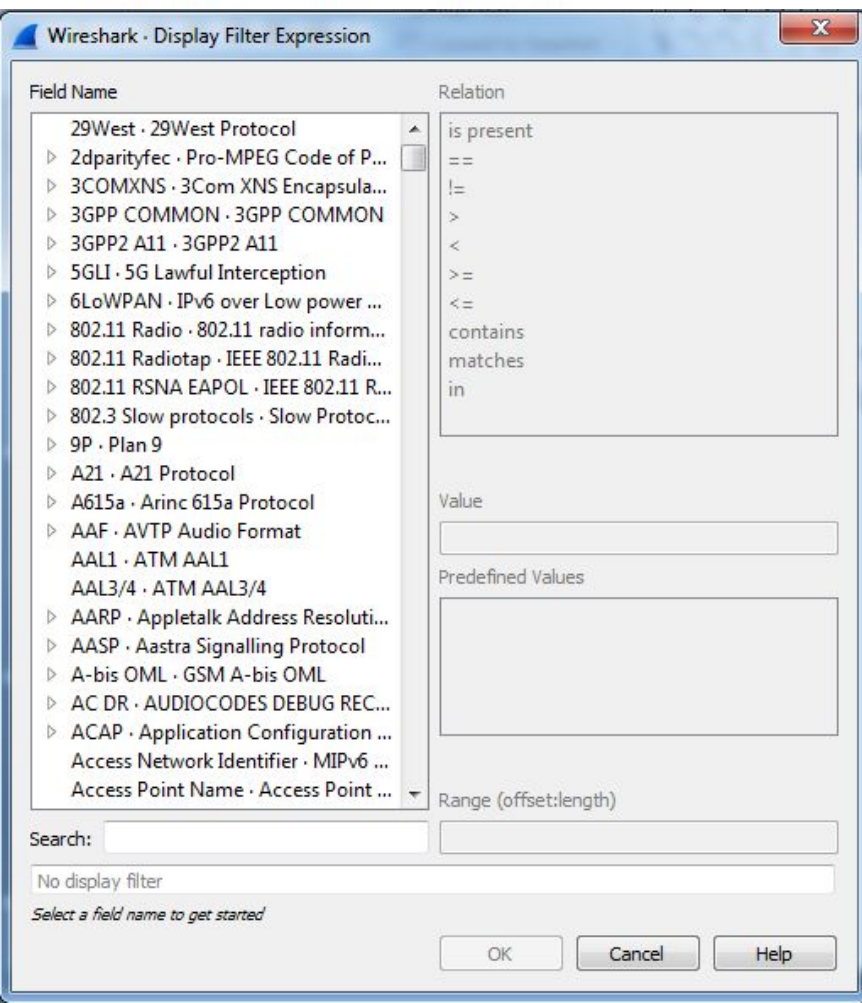
```
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) &&  
  (udp.port==67 || udp.port==68)
```

```
tcp.dstport == 80
```

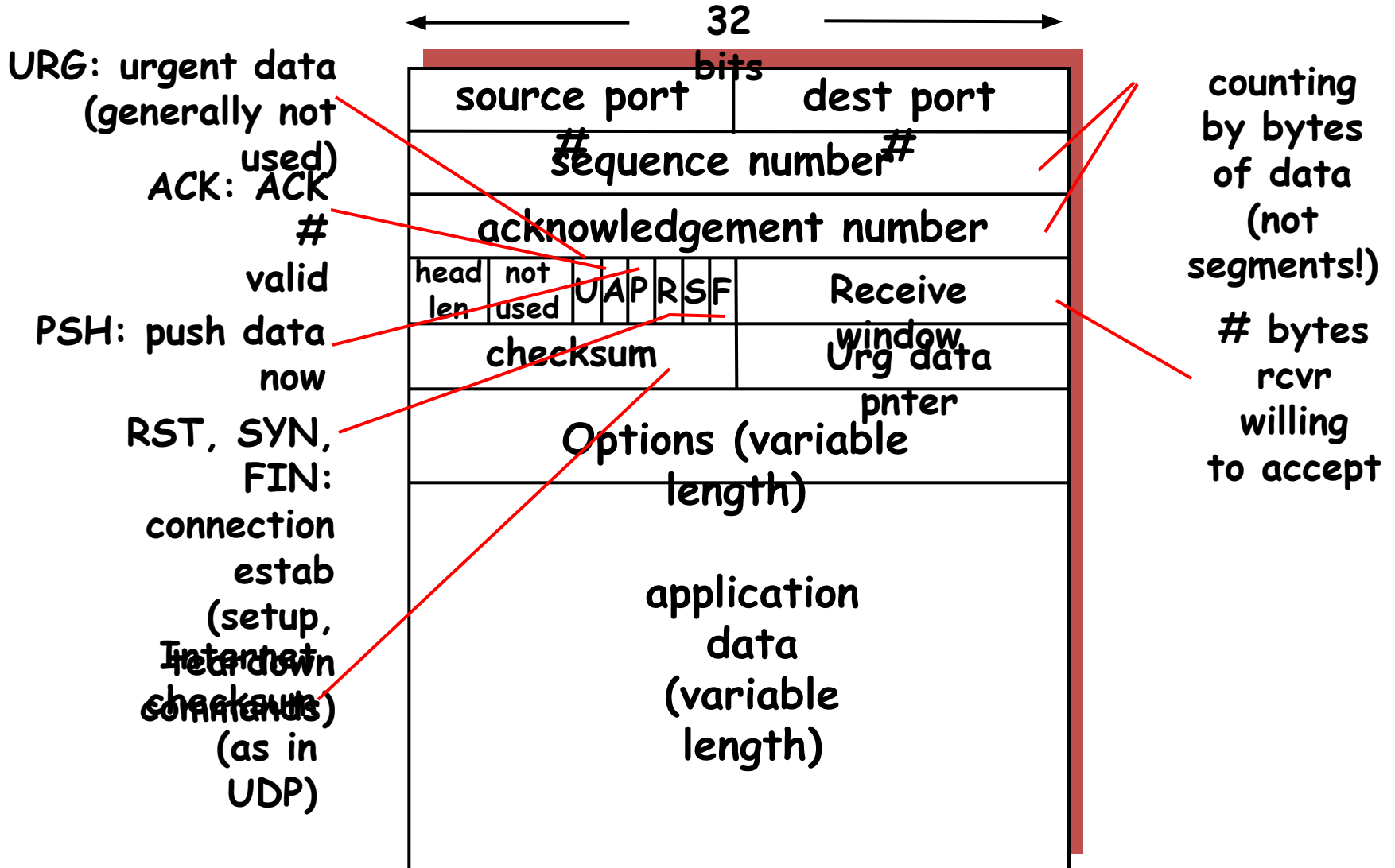
Display Filter

Syntax: **Protocol** . **String 1** . **String 2** **Comparison operator** **Value** **Logical Operations** **Other expression**

Example: ftp passive ip == 10.2.3.4 xor icmp.type



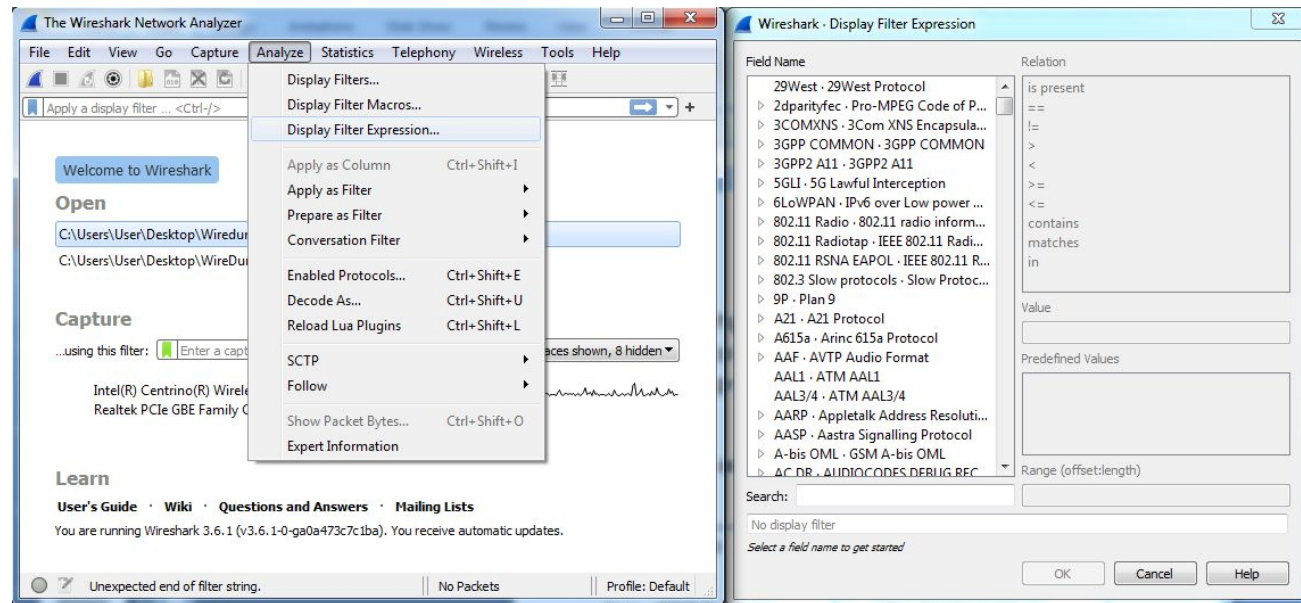
TCP Segment Structure



Display Filter

- String1, String2 (Optional settings):
 - Sub protocol categories inside the protocol.
 - Look for a protocol and then click on the "+" character.
 - Example:
 - **tcp.srcport == 80**
 - **tcp.flags == 2**
 - SYN packet
 - Tcp.flags.syn==1
 - **tcp.flags == 18**
 - SYN/ACK
 - **TCP Flag field:**

U	A	P	R	S	F
R	C	S	S	Y	I
G	K	H	T	N	N



Display Filter Expressions

Six **comparison** operators are available:

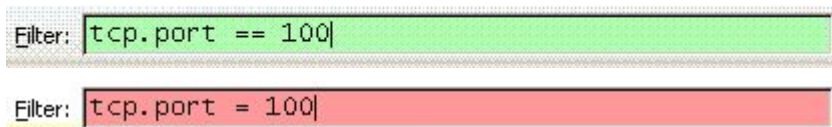
English format:	C like format:	Meaning:
eq	==	Equal
ne	!=	Not equal
gt	>	Greater than
lt	<	Less than
ge	>=	Greater or equal
le	<=	Less or equal

→ **Logical expressions:**

English format:	C like format:	Meaning:
and	&&	Logical AND
or		Logical OR
xor	^^	Logical XOR
not	!	Logical NOT

- `snmp || dns || icmp`
 - Display the SNMP or DNS or ICMP traffics.
- `tcp.port == 25`
 - Display packets with TCP source or destination port 25.
- `tcp.flags`
 - Display packets having a TCP flag.
- `tcp.flags == 0x02`
 - Display packets with a TCP SYN flag.

If the filter syntax is correct, it will be highlighted in green, otherwise if there is a syntax mistake it will be highlighted in red.



Correct syntax

Wrong syntax

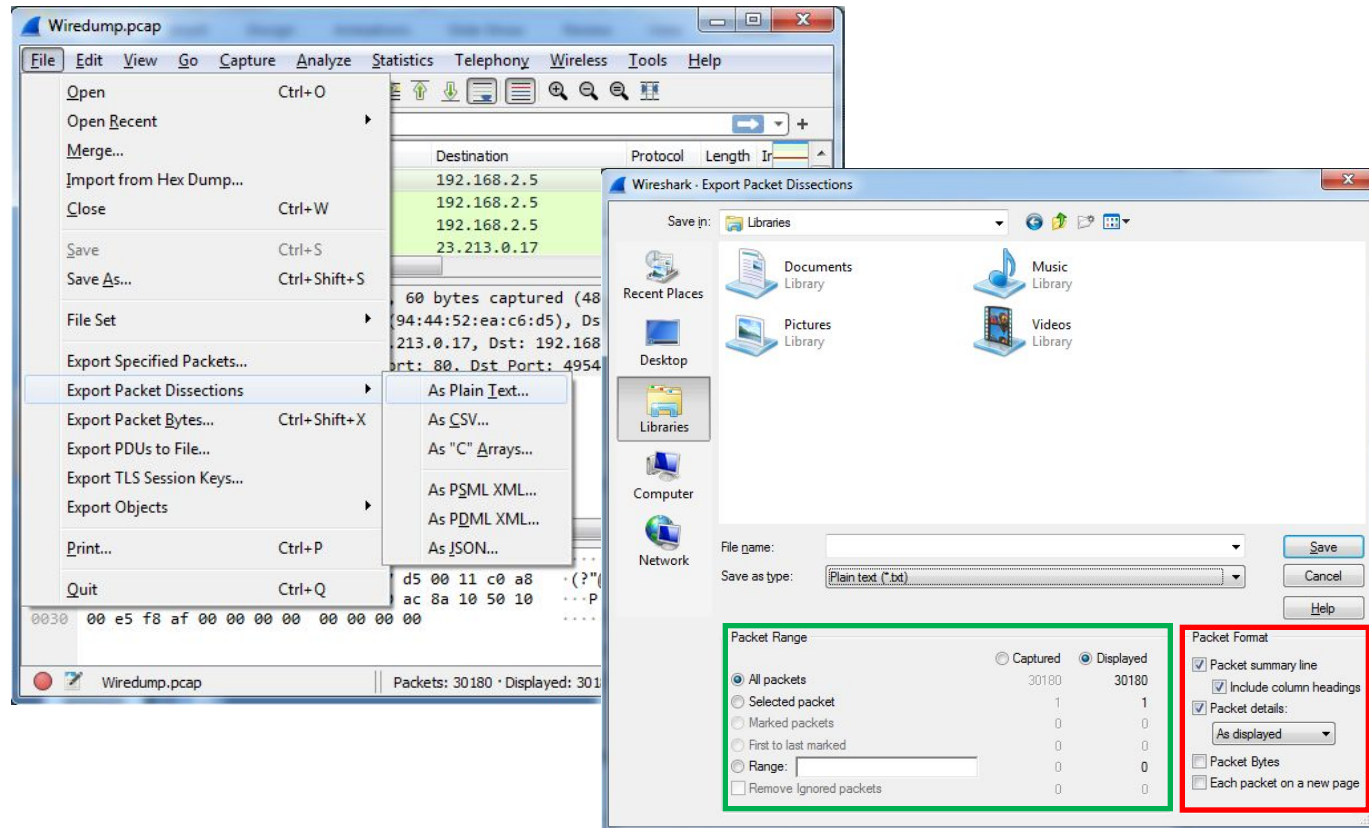
Save Filtered Packets after Using Display Filter

- We can also save all filtered packets in text file for further analysis
- Operation:

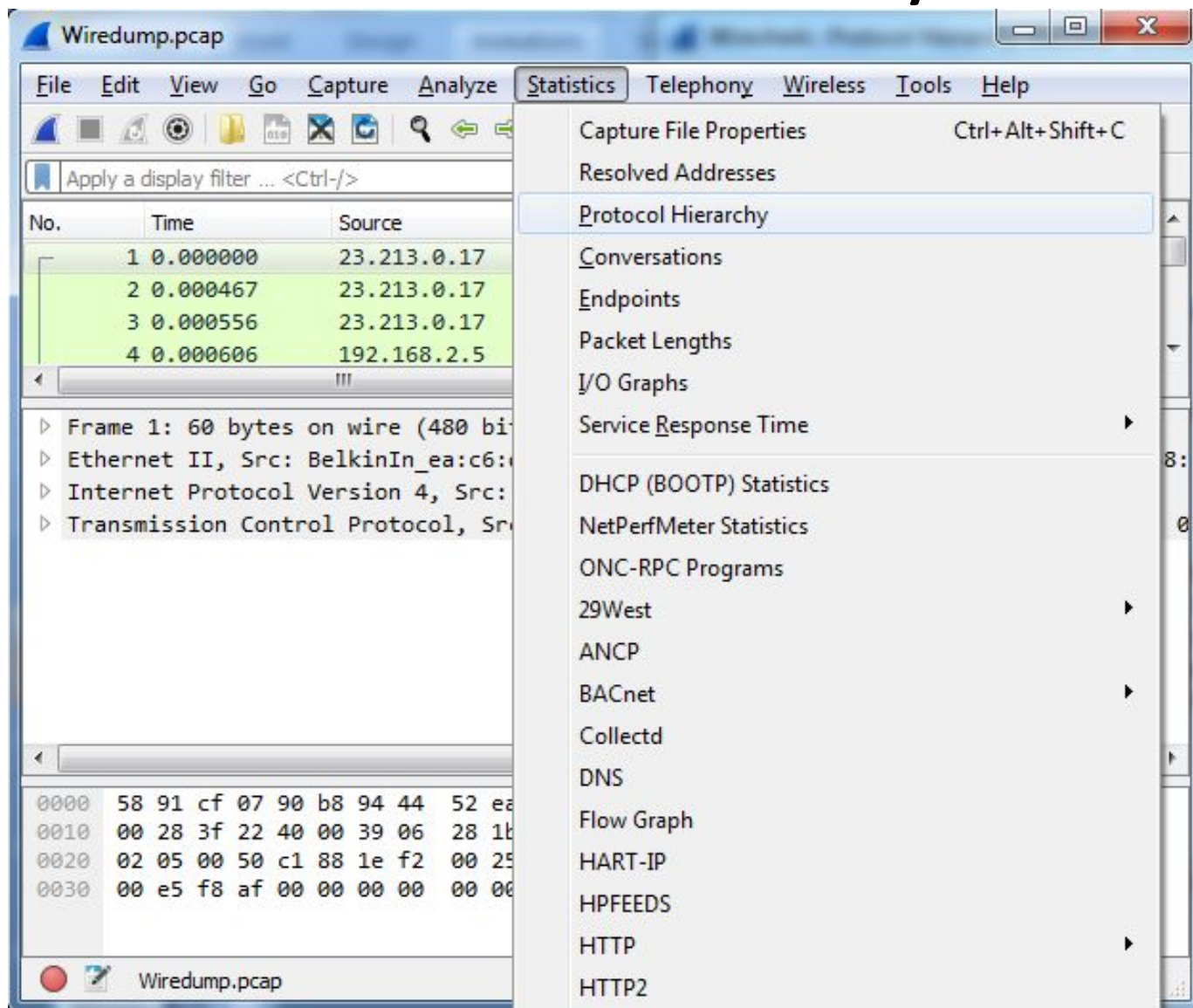
File ☐ Export packet dissections
☐ as “plain text” file

1). In “packet range” option, select “Displayed”

2). Choose “summary line” or “detail”



Protocol Hierarchy



Protocol Hierarchy (contd.)

Wireshark · Protocol Hierarchy Statistics · WireDump.pcap

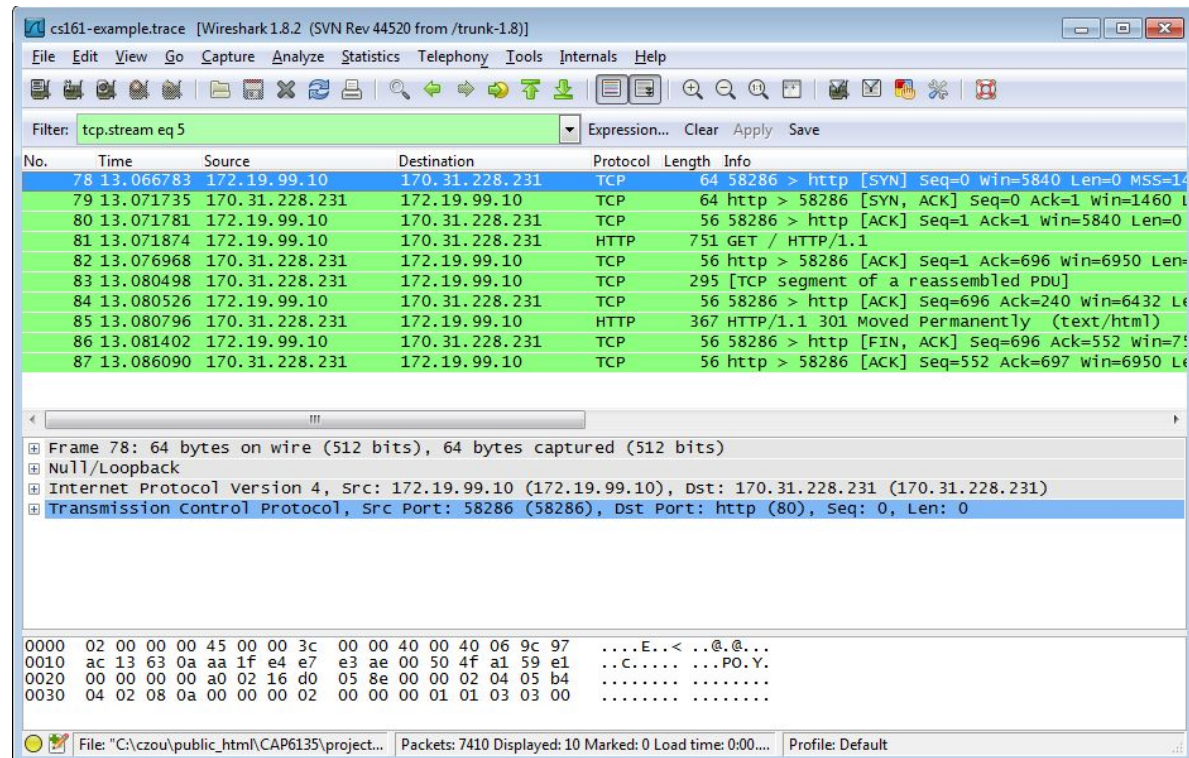
Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	30180	100.0	26368284	5445 k	0	0	0
Ethernet	100.0	30180	1.6	422520	87 k	0	0	0
Internet Protocol Version 6	0.0	9	0.0	360	74	0	0	0
User Datagram Protocol	0.0	3	0.0	24	4	0	0	0
DHCPv6	0.0	3	0.0	261	53	3	261	53
Internet Control Message Protocol v6	0.0	6	0.0	192	39	6	192	39
Internet Protocol Version 4	99.9	30160	2.3	603228	124 k	0	0	0
User Datagram Protocol	8.9	2691	0.1	21528	4445	0	0	0
Simple Service Discovery Protocol	0.2	63	0.1	16611	3430	63	16611	3430
QUIC IETF	8.3	2496	5.8	1517524	313 k	2439	1471948	303 k
Malformed Packet	0.0	2	0.0	0	0	2	0	0
NetBIOS Name Service	0.1	23	0.0	1240	256	23	1240	256
NetBIOS Datagram Service	0.0	4	0.0	765	157	0	0	0
SMB (Server Message Block Protocol)	0.0	4	0.0	437	90	0	0	0
SMB MailSlot Protocol	0.0	4	0.0	100	20	0	0	0
Microsoft Windows Browser Protocol	0.0	4	0.0	93	19	4	93	19
Multicast Domain Name System	0.0	4	0.0	244	50	4	244	50
Dynamic Host Configuration Protocol	0.0	3	0.0	900	185	3	900	185
Domain Name System	0.3	95	0.1	14561	3006	95	14561	3006
Data	0.2	58	0.0	2278	470	58	2278	470
Transmission Control Protocol	91.0	27457	90.2	23775639	4909 k	20838	15200882	3139 k
Transport Layer Security	22.1	6658	82.3	21701465	4481 k	6513	18459368	3812 k

No display filter.

Close Copy Help

Filter out/in Single TCP Stream

- When click “filter out this TCP stream” in previous page’s box, new filter string will contain like:
 - http and !(tcp.stream eq 5)
- So, if you use “tcp.stream eq 5” as filter string, you keep this HTTP session



Old version

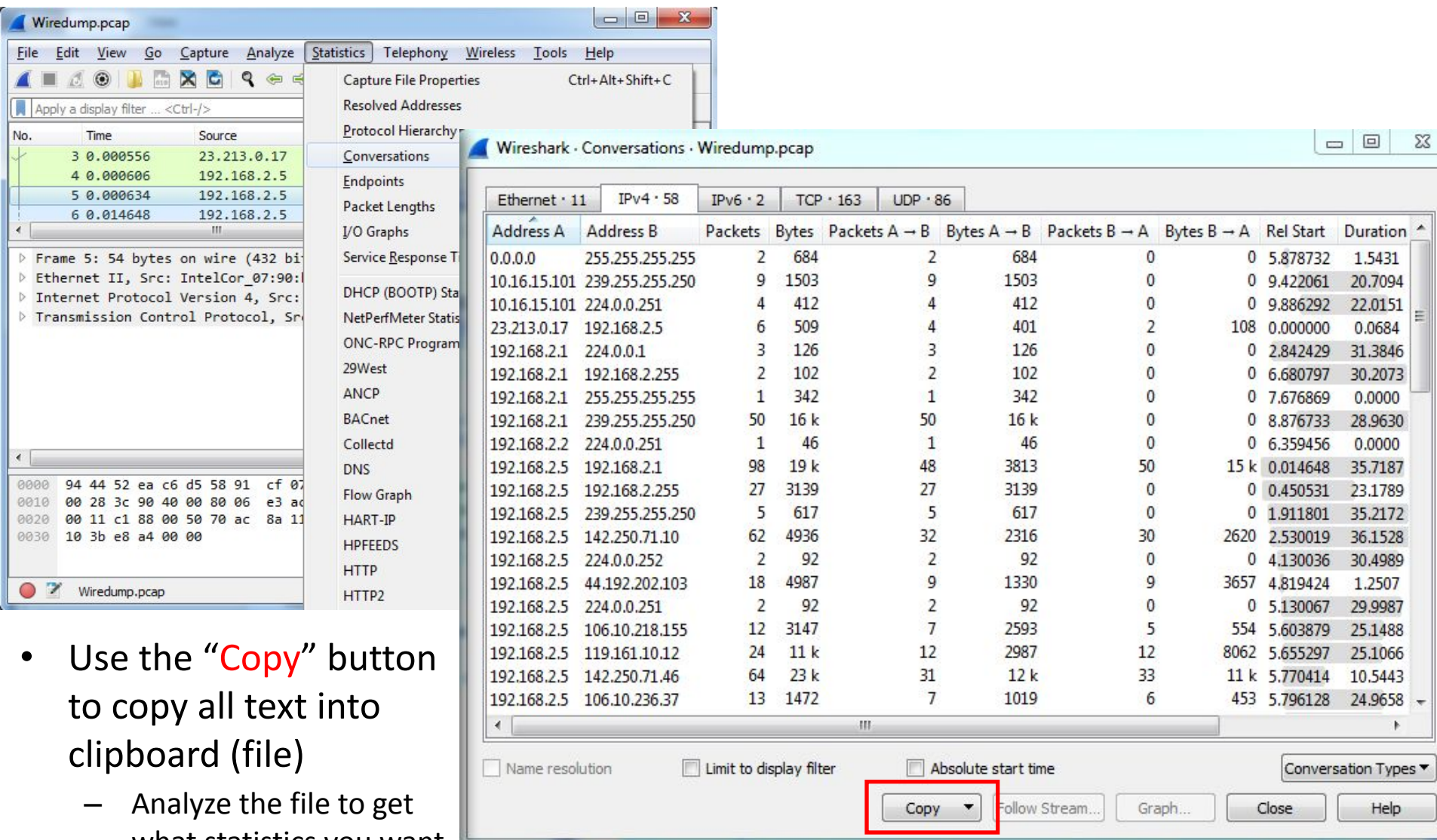
Expert Info

The image shows the Wireshark interface with the 'Expert Information' window open. The main window displays a packet capture of 'Wiredump.pcap'. The 'Expert Information' window lists various errors and warnings, categorized by severity (Error, Warning, Note) and grouped by protocol (TLS, DNS, TCP, QUIC, IPv4).

Severity	Summary	Group	Protocol
Error	TLSCiphertext length MUST NOT exceed $2^{14} + 2048$	Protocol	TLS
Error	Malformed Packet (Exception occurred)	Malformed	TLS
Error	Vector length 185 is too large, truncating it to 7	Malformed	TLS
Error	C:\gitlab-builds\builds\~fyeYoMP\1\wireshark\wireshark\...	Malformed	TLS
Warning	DNS query retransmission. Original request in frame 19250	Protocol	DNS
Warning	Connection reset (RST)	Sequence	TCP
Warning	ACKed segment that wasn't captured (common at capture...	Sequence	TCP
Warning	This frame is a (suspected) out-of-order segment	Sequence	TCP
Warning	Ignored Unknown Record	Protocol	TLS
Warning	TCP Zero Window segment	Sequence	TCP
Warning	TCP window specified by the receiver is now completely full	Sequence	TCP
Warning	Previous segment(s) not captured (common at capture sta...	Sequence	TCP
Warning	D-SACK Sequence	Sequence	TCP
Warning	Failed to create decryption context: Secrets are not available	Decryption	QUIC
Note	This frame is a (suspected) fast retransmission	Sequence	TCP
Note	"Time To Live" != 255 for a packet sent to the Local Netwo...	Sequence	IPv4
Note	Duplicate ACK (#1)	Sequence	TCP
Note	This frame is a (suspected) spurious retransmission	Sequence	TCP
Note	This frame is a (suspected) retransmission	Sequence	TCP
Note	This session reuses previously negotiated keys (Session res...	Sequence	TLS
Note	"Time To Live" only 4	Sequence	IPv4

At the bottom of the Expert Information window, there are checkboxes for 'Limit to Display Filter' (unchecked) and 'Group by summary' (checked). A search bar and 'Show...' button are also present. 'Close' and 'Help' buttons are at the bottom right.

Conversations

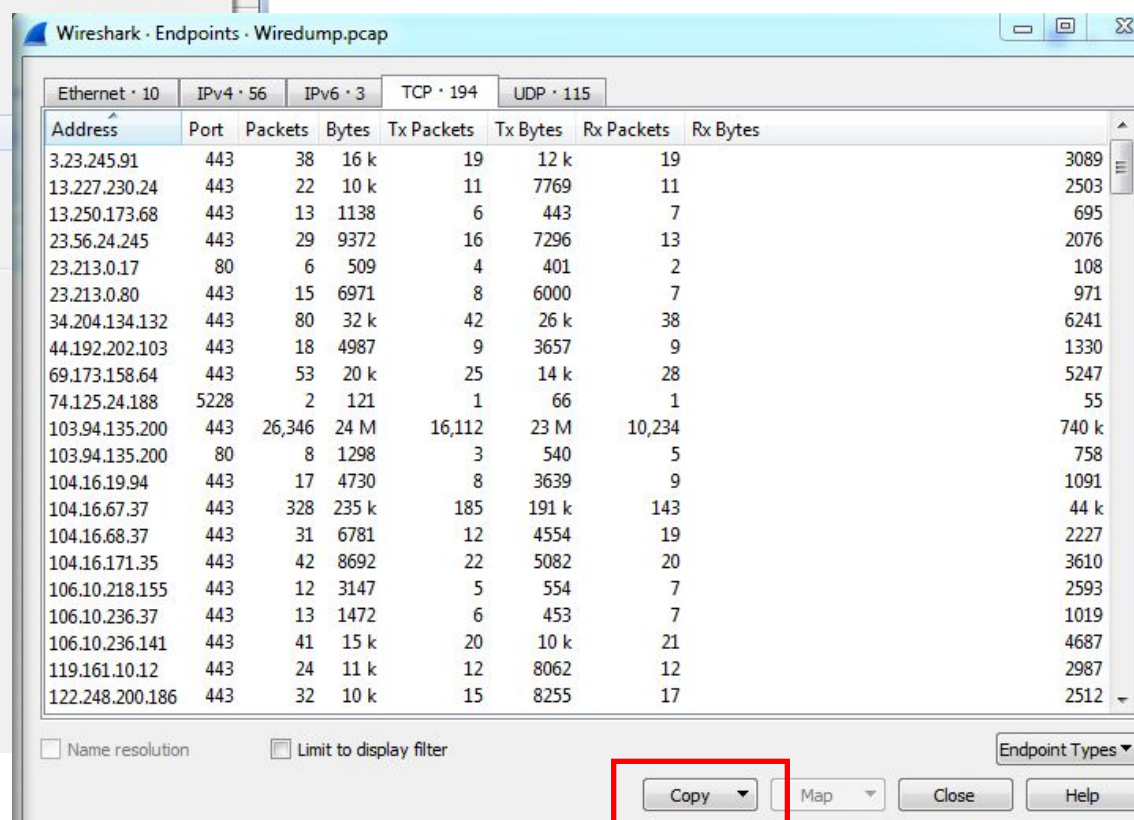


The screenshot shows the Wireshark interface with the 'Conversations' pane open. The 'Copy' button is highlighted with a red box. The 'Conversations' pane displays a table of network conversations with columns for Address A, Address B, Packets, Bytes, and various statistics.

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration
0.0.0.0	255.255.255.255	2	684	2	684	0	0	5.878732	1.5431
10.16.15.101	239.255.255.250	9	1503	9	1503	0	0	9.422061	20.7094
10.16.15.101	224.0.0.251	4	412	4	412	0	0	9.886292	22.0151
23.213.0.17	192.168.2.5	6	509	4	401	2	108	0.000000	0.0684
192.168.2.1	224.0.0.1	3	126	3	126	0	0	2.842429	31.3846
192.168.2.1	192.168.2.255	2	102	2	102	0	0	6.680797	30.2073
192.168.2.1	255.255.255.255	1	342	1	342	0	0	7.676869	0.0000
192.168.2.1	239.255.255.250	50	16 k	50	16 k	0	0	8.876733	28.9630
192.168.2.2	224.0.0.251	1	46	1	46	0	0	6.359456	0.0000
192.168.2.5	192.168.2.1	98	19 k	48	3813	50	15 k	0.014648	35.7187
192.168.2.5	192.168.2.255	27	3139	27	3139	0	0	0.450531	23.1789
192.168.2.5	239.255.255.250	5	617	5	617	0	0	1.911801	35.2172
192.168.2.5	142.250.71.10	62	4936	32	2316	30	2620	2.530019	36.1528
192.168.2.5	224.0.0.252	2	92	2	92	0	0	4.130036	30.4989
192.168.2.5	44.192.202.103	18	4987	9	1330	9	3657	4.819424	1.2507
192.168.2.5	224.0.0.251	2	92	2	92	0	0	5.130067	29.9987
192.168.2.5	106.10.218.155	12	3147	7	2593	5	554	5.603879	25.1488
192.168.2.5	119.161.10.12	24	11 k	12	2987	12	8062	5.655297	25.1066
192.168.2.5	142.250.71.46	64	23 k	31	12 k	33	11 k	5.770414	10.5443
192.168.2.5	106.10.236.37	13	1472	7	1019	6	453	5.796128	24.9658

- Use the “Copy” button to copy all text into clipboard (file)
 - Analyze the file to get what statistics you want

Find EndPoint Statistics



The image shows the Wireshark interface with the 'Endpoints' window open. The 'Endpoints' window displays a table of statistics for various network endpoints. The table has columns for Address, Port, Packets, Bytes, Tx Packets, Tx Bytes, Rx Packets, and Rx Bytes. The data is sorted by Rx Bytes in descending order. The 'Copy' button is highlighted with a red box.

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
3.23.245.91	443	38	16 k	19	12 k	19	3089
13.227.230.24	443	22	10 k	11	7769	11	2503
13.250.173.68	443	13	1138	6	443	7	695
23.56.24.245	443	29	9372	16	7296	13	2076
23.213.0.17	80	6	509	4	401	2	108
23.213.0.80	443	15	6971	8	6000	7	971
34.204.134.132	443	80	32 k	42	26 k	38	6241
44.192.202.103	443	18	4987	9	3657	9	1330
69.173.158.64	443	53	20 k	25	14 k	28	5247
74.125.24.188	5228	2	121	1	66	1	55
103.94.135.200	443	26,346	24 M	16,112	23 M	10,234	740 k
103.94.135.200	80	8	1298	3	540	5	758
104.16.19.94	443	17	4730	8	3639	9	1091
104.16.67.37	443	328	235 k	185	191 k	143	44 k
104.16.68.37	443	31	6781	12	4554	19	2227
104.16.171.35	443	42	8692	22	5082	20	3610
106.10.218.155	443	12	3147	5	554	7	2593
106.10.236.37	443	13	1472	6	453	7	1019
106.10.236.141	443	41	15 k	20	10 k	21	4687
119.161.10.12	443	24	11 k	12	8062	12	2987
122.248.200.186	443	32	10 k	15	8255	17	2512

- You can sort by field
- Use the “Copy” button to copy all text into clipboard (file)
 - Analyze the file to get what statistics you want

On-line Wireshark Trace Files

- Public available .pcap files:
 - <http://www.netresec.com/?page=PcapFiles>
- <http://www.tp.org/jay/nwanalysis/traces/Lab%20Trace%20Files/>
- Wiki Sample capture
 - <https://wiki.wireshark.org/SampleCaptures>

Example Trace File and Questions

- Network Forensic Puzzle Contests
 - <http://forensicscontest.com/2010/02/03/puzzle-4-the-curious-mr-x>
- SharkFest'15 Packet Challenge
 - <https://sharkfest.wireshark.org/assets/presentations15/packetchallenge.zip>

Source of Installer

- <https://www.wireshark.org/download.html>

Thank You

Acknowledgement: Web sources